Additional Material for Data-driven personalized neural adaptation for robust auditory attention decoding

1 Purpose and interpretation

This supplement reports the analyses added in response to peer review (PAL). The primary estimand is the within-backbone, within-subject change from the reproduced baseline to PAL under an identical experimental pipeline. Absolute results from different backbones are not interpreted as a causal ranking because the native implementations differ in data partitioning, reference construction, feature extraction, validation, and early stopping.

2 Published and reproduced baselines

Table 1 separates reproduction fidelity from the evaluation of PAL. For each dataset, backbone, and decision window, it reports the published or reference accuracy, the reproduced baseline, and the difference between them. These reference values are provided only for transparency and are not used to calculate PAL gains. The KUL reference values for SSF-CNN come from its original study [1], whereas the DTU values are the reproduction reported in the DBPNet study [2]. The STAnet values come from its original study [3]. For DBPNet and DARNet, we use the leakage-controlled DBPNet2 and DARNet2 evaluations reported in the DARNet study [4].

Table 1: Published/reference and reproduced baseline accuracy (mean $\pm$ standard deviation, %). The final column is reproduced minus published/reference in percentage points.

Dataset	Backbone	Window (s)	Published/ref.	Reproduced	Diff. (pp)
KUL	SSF-CNN	0.1	67.20 ± 4.57	61.71 ± 5.73	-5.5
KUL	SSF-CNN	1.0	81.70 ± 5.37	60.72 ± 6.59	-21.0
KUL	SSF-CNN	2.0	84.70 ± 6.13	59.16 ± 6.69	-25.5
KUL	STAnet	0.1	80.80 ± 9.87	83.64 ± 5.88	$+2.8$
KUL	STAnet	1.0	90.10 ± 8.95	92.32 ± 5.52	$+2.2$
KUL	STAnet	2.0	91.40 ± 8.22	91.26 ± 6.04	-0.1
KUL	DBPNet	0.1	85.30 ± 6.22	86.69 ± 9.34	$+1.4$
KUL	DBPNet	1.0	94.40 ± 4.62	92.96 ± 7.59	-1.4
KUL	DBPNet	2.0	95.30 ± 4.63	93.12 ± 7.81	-2.2
KUL	DARNet	0.1	89.20 ± 5.50	87.28 ± 10.20	-1.9
KUL	DARNet	1.0	94.80 ± 4.53	91.91 ± 8.72	-2.9
KUL	DARNet	2.0	95.50 ± 4.89	92.15 ± 8.41	-3.3
DTU	SSF-CNN	0.1	62.50 ± 3.40	63.92 ± 4.54	$+1.4$
DTU	SSF-CNN	1.0	69.80 ± 5.12	65.84 ± 5.05	-4.0
DTU	SSF-CNN	2.0	73.30 ± 6.21	67.94 ± 4.77	-5.4
DTU	STAnet	0.1	65.70 ± 5.50	59.09 ± 3.80	-6.6
DTU	STAnet	1.0	71.90 ± 8.94	64.12 ± 6.81	-7.8
DTU	STAnet	2.0	73.70 ± 9.59	61.59 ± 5.64	-12.1
DTU	DBPNet	0.1	74.00 ± 5.20	70.72 ± 5.24	-3.3
DTU	DBPNet	1.0	79.80 ± 6.91	77.55 ± 4.80	-2.3
DTU	DBPNet	2.0	80.20 ± 6.79	80.88 ± 6.39	$+0.7$
DTU	DARNet	0.1	74.60 ± 6.09	73.94 ± 5.29	-0.7
DTU	DARNet	1.0	80.10 ± 6.85	78.90 ± 5.60	-1.2
DTU	DARNet	2.0	81.20 ± 6.34	80.59 ± 5.24	-0.6

For DBPNet and DARNet, most reproduced values differ from their references by approximately three percentage points or less. Random initialization, partition construction, checkpoint selection, early stopping, and other implementation details may contribute to these differences. In the present pipeline, CSP is fitted only on the training portion and is then applied to the validation and test portions. Thus, each PAL result and its matched baseline follow the same leakage-controlled data path.

The largest KUL discrepancy occurs for SSF-CNN at the 1- and 2-s windows: the reference accuracies are 81.7% and 84.7%, whereas the reproduced accuracies are 60.72% and 59.16%. SSF-CNN depends on window partitioning, alpha-power-map construction, two-dimensional scalp interpolation, normalization, and training details. Its source paper specifies an 80%/10%/10% split and 50% window overlap but does not fully specify the random partition, topographic interpolation, or training seed. In our implementation, continuous EEG is first divided into training, validation, and test segments; overlapping windows are then generated separately within each segment. This split-before-windowing procedure prevents closely neighboring windows from crossing subset boundaries and reduces the number of highly similar training examples, particularly for longer windows. We therefore describe the result as a protocol-dependent reproduction rather than an exact replication.

The other prominent discrepancy is STAnet on DTU. The reference accuracies at 0.1, 1, and 2 s are 65.7%, 71.9%, and 73.7%, compared with 59.09%, 64.12%, and 61.59% in our reproduction. DTU includes multiple event codes, acoustic conditions, and segments that are not standard two-speaker trials, while the original method description does not fully specify event-code mapping, trial boundaries, cropping, channel order, or filter parameters. We reconstruct two-speaker trials from the event information, retain the first 64 scalp channels, exclude non-two-speaker trials, and centrally crop each nominal 50-s trial to 48 s to match the reported sample count. These decisions affect DTU more strongly than KUL. Their influence also increases at longer windows because each trial yields fewer independent samples.

Every PAL gain in the manuscript is calculated relative to the corresponding reproduced baseline using the same code path, preprocessing, subjects, split, training settings, and metric. The published or reference values serve only to identify close reproductions and protocol-dependent gaps. Accordingly, the analyses below treat paired within-backbone changes as the primary evidence and use the native-versus-unified comparison to show how strongly those changes depend on the evaluation protocol.

3 Computational cost and parameter overhead of PAL

Table 2 quantifies the additional parameters, floating-point operations, inference latency, and training-step time introduced by PAL. Each row corresponds to the rank selected for that specific backbone–dataset–window configuration; values are not averaged across datasets or decision windows. The purpose is to define "lightweight" in measurable terms, not to imply zero training or inference cost.

Table 2: Computational overhead of PAL for every backbone–dataset–window configuration. "Params" denotes trainable parameters; latency and training-step times are in milliseconds; FLOPs are cached-matrix PAL FLOPs in millions.

Backbone	Data	Window	r	Base params	PAL params	Added	Added (%)	PAL FLOPs	Base inf.	PAL inf.	Δ inf.	Base step	PAL step	Δ step
DARNet	DTU	0.1	64	69674	77994	8320	11.94	0.058	1.706	1.869	0.163	8.259	8.983	0.724
DARNet	DTU	1.0	64	69674	77994	8320	11.94	0.528	1.529	1.721	0.192	7.729	8.803	1.075
DARNet	DTU	2.0	64	69674	77994	8320	11.94	1.057	1.519	1.866	0.347	6.718	8.982	1.364
DARNet	KUL	0.1	32	69674	73898	4224	6.06	0.107	1.967	1.700	-0.267	9.397	10.175	0.778
DARNet	KUL	1.0	8	69674	70826	1152	1.65	0.057	1.689	2.030	0.341	7.909	8.608	0.699
DARNet	KUL	2.0	4	69674	70314	640	0.92	0.114	1.777	2.135	0.358	7.605	8.825	1.220
DBPNet	DTU	0.1	64	915154	923474	8320	0.91	0.107	3.630	3.964	0.335	21.528	26.831	5.303
DBPNet	DTU	1.0	64	915154	923474	8320	0.91	0.514	3.497	4.053	0.557	18.523	20.249	1.726
DBPNet	DTU	2.0	64	915154	923474	8320	0.91	1.057	3.594	4.025	0.431	14.985	19.547	4.562
DBPNet	KUL	0.1	32	915154	919378	4224	0.46	0.107	4.685	3.986	-0.699	15.854	19.786	3.932
DBPNet	KUL	1.0	8	915154	916306	1152	0.13	0.057	3.592	4.017	0.425	13.380	19.837	6.453
DBPNet	KUL	2.0	4	915154	915794	640	0.07	0.114	3.632	4.261	0.629	19.787	26.890	7.103
SSF-CNN	DTU	0.1	64	4211682	4220002	8320	0.20	0.058	0.616	0.649	0.033	11.317	12.998	1.682
SSF-CNN	DTU	1.0	64	4211682	4220002	8320	0.20	0.528	0.720	0.772	0.051	14.126	16.612	2.486
SSF-CNN	DTU	2.0	64	4211682	4220002	8320	0.20	1.057	0.751	0.784	0.034	2.584	13.297	10.713
SSF-CNN	KUL	0.1	32	4211682	4215906	4224	0.10	0.107	0.587	0.636	0.049	2.420	13.825	10.405
SSF-CNN	KUL	1.0	8	4211682	4212834	1152	0.03	0.057	0.730	0.758	0.028	3.300	13.556	10.257
SSF-CNN	KUL	2.0	4	4211682	4212322	640	0.02	0.114	0.732	0.787	0.055	11.349	13.044	1.695
STAnet	DTU	0.1	64	3505	9825	6320	180.19	0.058	0.470	0.513	0.043	2.450	3.043	0.591
STAnet	DTU	1.0	64	3910	12230	8320	212.79	0.528	0.863	0.865	0.003	5.568	5.820	0.253
STAnet	DTU	2.0	64	4358	12678	8320	190.91	1.114	1.017	1.139	0.123	8.972	9.905	0.934
STAnet	KUL	0.1	32	3505	7729	4224	120.51	0.107	0.417	0.535	0.118	2.295	2.449	0.154
STAnet	KUL	1.0	8	3910	5062	1152	29.46	0.057	0.806	0.832	0.025	5.719	6.208	0.490
STAnet	KUL	2.0	4	4358	4998	640	14.69	0.114	0.992	1.068	0.076	8.806	9.007	0.201

PAL adds one front-end transformation while leaving the internal architecture, feature branches, and classifier of STAnet, SSF-CNN, DBPNet, and DARNet unchanged. REST deterministically maps the signal to a standard-head-model reference space, after which the personalized residual matrix acts before backbone-specific feature extraction. Thus, "plug-and-play" refers to implementation compatibility: PAL can be inserted or removed without redesigning the downstream network. It does not imply equal performance gains for every backbone.

The selected ranks in Table 2 range from $r = 4$ to $r = 64$. Selection used validation performance only; test accuracy, test checkpoints, and other test-set statistics were excluded. The separate $r = 2$–64 sweep in the sensitivity analysis indicates broad plateaus for most backbones, while also identifying the stronger sensitivity of SSF-CNN. PAL and its backbone are optimized jointly with the stated optimizer and schedule, without a second optimization stage or an auxiliary feature-extraction network.

For a configuration with C input channels and rank r, the modulation contributes $2Cr+2C$ trainable parameters. The configurations in Table 2 add 640–8,320 parameters. STAnet shows the largest relative increases because its baseline has only a few thousand parameters; the maximum relative increases for DBPNet, DARNet, and SSF-CNN are 0.91%, 11.94%, and 0.20%, respectively. Absolute and relative parameter counts should therefore be interpreted together.

At inference, U, V, and the channel-scaling vector are pre-composed into the effective matrix W, so the low-rank factors need not be multiplied separately for every input. Under this cached-matrix implementation, the added cost in Table 2 is 0.058–1.114 MFLOPs and grows with the number of temporal samples. Latency was measured with batch size one after 100 warm-up runs, using 1,000 timed runs in each of five repetitions. The additional inference latency was generally below 0.63 ms. Slightly negative differences in several rows are treated as millisecond-scale measurement variation, not as acceleration caused by PAL.

Training-step timing covers the forward pass, loss computation, backward pass, and optimizer update. The largest added time in the table is 10.713 ms. This measured overhead is consistent with adding one input-stage spatial transformation and no iterative auxiliary procedure. The low-rank parameter count, cached-matrix inference cost, measured latency, and unchanged downstream backbone jointly support the qualified description of PAL as lightweight under the evaluated configurations.

4 Subject-level statistical analysis with multiplicity and dependence controls

The statistical unit is the subject. Within each configuration, folds or repeated runs are first averaged for each subject, and two-sided paired tests then compare baseline and PAL results from the same participants ($n = 16$ for KUL and $n = 18$ for DTU). Holm correction controls the family-wise error rate across all 24 primary backbone–dataset–window comparisons. Table 3 reports the mean paired gain, 95% confidence interval, paired Cohen's d_z, number of subjects improved, raw p-value, and Holm-adjusted p-value.

PAL produced a positive mean gain in all 24 native-protocol configurations. Fourteen comparisons remained significant after Holm correction, 19 had a mean gain of at least one percentage point, and 13 met both the Holm criterion and a 95% confidence-interval lower bound above one percentage point. At the individual level, 329 of the 408 subject–configuration pairs (80.6%) had a positive PAL-minus-baseline difference.

Because each subject contributes results for several backbones and windows, Table 4 provides two additional analyses using subject within dataset as the dependence unit (34 clusters). A linear mixed-effects model with dataset-specific subject random intercepts estimated an overall gain of 10.65 percentage points (95% CI [9.35, 11.96], $p = 1.436 \times 10^{-57}$). A cluster-robust linear model gave the same point estimate (95% CI [9.49, 11.81], $p = 5.613 \times 10^{-72}$). Agreement between these analyses shows that the overall estimate is not driven by treating a subject's repeated observations as independent.

Statistical significance and practical magnitude are interpreted separately. AAD has no universally accepted minimum important accuracy difference, so one percentage point is used only as a transparent descriptive threshold; the continuous gain, confidence interval, effect size, and subject-level improvement rate remain the primary evidence. The largest effects include DTU–STAnet at 0.1 s (37.96 points; 18/18 subjects improved), KUL–SSF-CNN at 2 s (29.73 points), and KUL–DBPNet at 0.1 s (11.11 points; 16/16

Table 3: Complete subject-level paired comparisons of PAL versus baseline across the 24 primary model–dataset–window settings. Accuracy values are mean ± subject-level standard deviation (%); gain and confidence intervals are percentage points. The raw paired-test p-values and Holm-adjusted p-values are calculated across the complete family of 24 comparisons.

Dataset	Model	Window	Baseline	PAL	Gain	95% CI	d_z	Improved	Raw p	Holm p
KUL	SSF-CNN	0.1 s	61.71 ± 5.73	81.37 ± 9.69	19.66	[14.70, 24.62]	2.11	16/16 (100.0%)	4.321×10^{-7}	7.778×10^{-6}
KUL	SSF-CNN	1 s	60.72 ± 6.59	88.19 ± 13.24	27.47	[21.11, 33.84]	2.30	15/16 (93.8%)	1.485×10^{-7}	2.970×10^{-6}
KUL	SSF-CNN	2 s	59.16 ± 6.69	88.89 ± 13.28	29.73	[23.56, 35.90]	2.57	15/16 (93.8%)	3.508×10^{-8}	7.367×10^{-7}
KUL	STAnet	0.1 s	83.64 ± 5.88	92.02 ± 4.49	8.38	[5.83, 10.94]	1.75	16/16 (100.0%)	4.339×10^{-6}	7.376×10^{-5}
KUL	STAnet	1 s	92.32 ± 5.52	94.68 ± 3.25	2.36	[0.21, 4.51]	0.58	10/16 (62.5%)	3.372×10^{-2}	2.361×10^{-1}
KUL	STAnet	2 s	91.26 ± 6.04	93.04 ± 4.18	1.78	[-0.63, 4.18]	0.39	9/16 (56.2%)	1.364×10^{-1}	8.183×10^{-1}
KUL	DBPNet	0.1 s	86.69 ± 9.34	97.80 ± 2.65	11.11	[5.81, 16.41]	1.12	16/16 (100.0%)	4.510×10^{-4}	6.765×10^{-3}
KUL	DBPNet	1 s	92.96 ± 7.59	98.37 ± 2.24	5.41	[1.70, 9.13]	0.78	13/16 (81.2%)	7.222×10^{-3}	7.222×10^{-2}
KUL	DBPNet	2 s	93.12 ± 7.81	98.12 ± 2.89	5.00	[1.18, 8.82]	0.70	13/16 (81.2%)	1.377×10^{-2}	1.239×10^{-1}
KUL	DARNet	0.1 s	87.28 ± 10.20	87.39 ± 9.61	0.10	[-0.63, 0.84]	0.07	6/16 (37.5%)	7.698×10^{-1}	1.000
KUL	DARNet	1 s	91.91 ± 8.72	92.22 ± 8.64	0.31	[-0.48, 1.11]	0.21	10/16 (62.5%)	4.149×10^{-1}	1.000
KUL	DARNet	2 s	92.15 ± 8.41	92.65 ± 7.67	0.50	[-0.62, 1.62]	0.24	7/16 (43.8%)	3.589×10^{-1}	1.000
DTU	SSF-CNN	0.1 s	63.92 ± 4.54	69.25 ± 7.37	5.33	[2.74, 7.93]	1.02	16/18 (88.9%)	4.529×10^{-4}	6.765×10^{-3}
DTU	SSF-CNN	1 s	65.84 ± 5.05	70.95 ± 8.71	5.11	[2.03, 8.19]	0.83	13/18 (72.2%)	2.722×10^{-3}	3.237×10^{-2}
DTU	SSF-CNN	2 s	67.94 ± 4.77	72.09 ± 8.19	4.15	[0.76, 7.53]	0.61	13/18 (72.2%)	1.923×10^{-2}	1.538×10^{-1}
DTU	STAnet	0.1 s	59.09 ± 3.80	97.05 ± 1.57	37.96	[35.81, 40.11]	8.79	18/18 (100.0%)	9.460×10^{-18}	2.270×10^{-16}
DTU	STAnet	1 s	64.12 ± 6.81	96.71 ± 3.21	32.58	[28.92, 36.24]	4.43	18/18 (100.0%)	8.304×10^{-13}	1.910×10^{-11}
DTU	STAnet	2 s	61.59 ± 5.64	94.03 ± 5.34	32.44	[28.79, 36.08]	4.42	18/18 (100.0%)	8.427×10^{-13}	1.910×10^{-11}
DTU	DBPNet	0.1 s	70.72 ± 5.24	78.10 ± 8.26	7.38	[3.48, 11.27]	0.94	12/18 (66.7%)	9.339×10^{-4}	1.214×10^{-2}
DTU	DBPNet	1 s	77.55 ± 4.80	86.18 ± 6.48	8.63	[6.38, 10.88]	1.91	18/18 (100.0%)	3.147×10^{-7}	5.979×10^{-6}
DTU	DBPNet	2 s	80.88 ± 6.39	86.81 ± 7.29	5.93	[3.92, 7.93]	1.47	18/18 (100.0%)	9.230×10^{-6}	1.477×10^{-4}
DTU	DARNet	0.1 s	73.94 ± 5.29	74.31 ± 5.66	0.37	[-0.13, 0.88]	0.37	11/18 (61.1%)	1.380×10^{-1}	8.183×10^{-1}
DTU	DARNet	1 s	78.90 ± 5.60	80.38 ± 6.44	1.48	[0.59, 2.37]	0.83	15/18 (83.3%)	2.697×10^{-3}	3.237×10^{-2}
DTU	DARNet	2 s	80.59 ± 5.24	81.24 ± 6.04	0.66	[-0.28, 1.60]	0.35	13/18 (72.2%)	1.587×10^{-1}	8.183×10^{-1}

Table 4: Sensitivity analyses accounting for within-subject dependence in PAL–baseline accuracy differences.

Method	Dependence handling	Observations	Clusters	Overall gain (pp)
Linear mixed-effects model	Dataset-specific subject random intercept	408	34	10.6503
Cluster-robust linear model	SE clustered by dataset-specific subject	408	34	10.6503

Method	95% CI	p-value
Linear mixed-effects model	[9.35, 11.96]	1.436×10^{-57}
Cluster-robust linear model	[9.49, 11.81]	5.613×10^{-72}

subjects improved). By contrast, the DARNet gains of 0.10–1.48 points are small, and most are not significant after Holm correction. The results therefore support an architecture-dependent effect rather than a universal improvement claim.

5 Statistical tests for the unified-control analysis

The unified-control sensitivity analysis uses the same subject-level procedure: a two-sided paired t-test for each backbone–dataset–window condition and a global Holm correction across the 24 contrasts. Table 5 reports each raw and adjusted p-value as raw/Holm.

Six contrasts remained significant after correction: DTU–DBPNet at 1 and 2 s; KUL–DBPNet at 0.1 s; KUL–STAnet at 0.1 s; and KUL–SSF-CNN at 1 and 2 s. The remaining contrasts do not provide multiplicity-adjusted evidence of a nonzero effect under the unified protocol. This more selective pattern is consistent with the architecture- and dataset-dependent gains reported for the unified-control experiment in the main manuscript.

Table 5: Paired significance tests for the unified-control experiment. Each cell reports the raw paired-t p-value / globally Holm-adjusted p-value across the 24 contrasts.

Dataset	Backbone	0.1 s	1 s	2 s
DTU	DARNet	0.1770/1.0000	0.7603/1.0000	0.8729/1.0000
	DBPNet	0.2828/1.0000	$7.927 \times 10^{-5}/1.823 \times 10^{-3}$	$1.087 \times 10^{-3}/0.02065$
	STAnet	0.6685/1.0000	0.8112/1.0000	0.2761/1.0000
	SSF-CNN	0.3821/1.0000	0.7594/1.0000	0.7270/1.0000
KUL	DARNet	0.7977/1.0000	0.2297/1.0000	0.4668/1.0000
	DBPNet	$2.743 \times 10^{-4}/5.760 \times 10^{-3}$	0.01541/0.26196	0.01763/0.28201
	STAnet	$8.053 \times 10^{-5}/1.823 \times 10^{-3}$	0.02342/0.35124	0.4595/1.0000
	SSF-CNN	0.01325/0.23848	$1.461 \times 10^{-5}/3.506 \times 10^{-4}$	$3.683 \times 10^{-4}/7.367 \times 10^{-3}$

6 Native-protocol versus unified-protocol PAL gains

Table 6 places the two evaluation roles side by side. The native analysis preserves each backbone's original preprocessing, partitioning, and optimization choices and is the primary test of PAL as a minimally invasive front end. The unified analysis holds data allocation and training controls in common and serves as a protocol-sensitivity check. Because the analyses estimate effects under different protocols, their gains should neither be pooled nor treated as direct replications of one another.

DTU–STAnet shows the largest protocol difference. Its native window-level gains are +37.96, +32.58, and +32.44 percentage points, whereas its unified trial-level gains are −0.88, +0.67, and +2.68 points. The protocols also differ in split unit, input distribution, randomization, and checkpoint selection, so the comparison does not isolate a single cause for this change. KUL–DBPNet and KUL–STAnet retain broadly similar gain profiles across protocols, and KUL–SSF-CNN retains large gains. Taken together, the results show that the estimated benefit of PAL depends on the backbone, dataset, decision window, and evaluation protocol.

Table 6: PAL–baseline gains (percentage points) under the native and unified-control protocols. Columns give gains at the 0.1 s, 1 s, and 2 s decision windows for each protocol.

		Native protocol			Unified control		
Dataset	Backbone	0.1 s	1 s	2 s	0.1 s	1 s	2 s
DTU	DARNet	+0.37	+1.48	+0.66	+1.26	+0.47	+0.39
	DBPNet	+7.38	+8.63	+5.93	+0.97	+6.11	+4.44
	STAnet	+37.96	+32.58	+32.44	-0.88	+0.67	+2.68
	SSF-CNN	+5.33	+5.11	+4.15	+1.29	+0.54	-0.88
KUL	DARNet	+0.10	+0.31	+0.50	-0.12	+0.78	+0.81
	DBPNet	+11.11	+5.41	+5.00	+11.62	+5.26	+4.08
	STAnet	+8.38	+2.36	+1.78	+8.23	+2.21	+0.76
	SSF-CNN	+19.66	+27.47	+29.73	+15.84	+30.73	+25.77

7 Train, validation, and test performance

This analysis separates training, validation, and test behavior to determine whether the reported test gains merely reflect greater fitting of the training data. Table 7 reports the complete partition-level records for all 24 dataset–model–window settings; Table 8 then summarizes the PAL-minus-baseline change for the 12 settings with complete validation records. SSF-CNN improves by similar amounts across the three partitions, whereas DARNet has nearly unchanged training accuracy and smaller validation and test gains.

Table 7: Complete train, validation, and test performance for every native-protocol dataset–model–window setting. Losses are means; accuracies and gaps are percentages or percentage points. Validation entries are NA for native STAnet and DBPNet, which do not define an independent validation partition. Gain columns are reported on the matched baseline row.

Dataset	Model	Window	Condition	Train Loss Mean	Train Acc Mean (%)	Valid Loss Mean	Valid Acc Mean (%)	Test Loss Mean	Test Acc Mean (%)	Train–Test Gap (pp)	Abs Valid–Test Gap (pp)	Train Acc Gain vs Baseline (pp)	Valid Acc Gain vs Baseline (pp)	Test Acc Gain vs Baseline (pp)
KUL	SSF-CNN	0.1	Baseline	0.6054	65.72	0.6637	60.88	0.6585	61.71	4.0100	0.8300	19.9300	20.3500	19.6600
KUL	SSF-CNN	0.1	PAL	0.3146	85.65	0.4617	81.23	0.4335	81.37	4.2800	0.1400	NA	NA	NA
KUL	SSF-CNN	1	Baseline	0.6146	64.48	0.6784	58.81	0.6897	60.72	3.7600	1.9100	27.4700	27.3100	27.4700
KUL	SSF-CNN	1	PAL	0.1639	91.95	0.4091	86.12	0.5598	88.19	3.7600	2.0700	NA	NA	NA
KUL	SSF-CNN	2	Baseline	0.6457	61.68	0.6917	57.07	0.6879	59.16	2.5200	2.0900	31.1900	30.7300	29.7300
KUL	SSF-CNN	2	PAL	0.1382	92.87	0.3161	87.80	0.4380	88.89	3.9800	1.0900	NA	NA	NA
KUL	STAnet	0.1	Baseline	0.3756	83.57	NA	NA	0.3668	83.64	-0.0700	NA	12.1900	NA	8.3800
KUL	STAnet	0.1	PAL	0.1392	95.76	NA	NA	0.2834	92.02	3.7400	NA	NA	NA	NA
KUL	STAnet	1	Baseline	0.1301	94.73	NA	NA	0.2372	92.32	2.4100	NA	3.5500	NA	2.3600
KUL	STAnet	1	PAL	0.0530	98.28	NA	NA	0.2575	94.68	3.6000	NA	NA	NA	NA
KUL	STAnet	2	Baseline	0.1029	95.95	NA	NA	0.3029	91.26	4.6900	NA	1.9100	NA	1.7800
KUL	STAnet	2	PAL	0.0602	97.86	NA	NA	0.3055	93.04	4.8200	NA	NA	NA	NA
KUL	DBPNet	0.1	Baseline	0.0467	98.27	NA	NA	0.6194	86.69	11.5800	NA	1.4100	NA	11.1100
KUL	DBPNet	0.1	PAL	0.0094	99.68	NA	NA	0.0805	97.80	1.8800	NA	NA	NA	NA
KUL	DBPNet	1	Baseline	0.0025	99.96	NA	NA	0.3012	92.96	7.0000	NA	0.0400	NA	5.4100
KUL	DBPNet	1	PAL	0.0004	100.00	NA	NA	0.0739	98.37	1.6300	NA	NA	NA	NA
KUL	DBPNet	2	Baseline	0.0036	99.99	NA	NA	0.2576	93.12	6.8700	NA	0.0100	NA	5.0000
KUL	DBPNet	2	PAL	0.0013	100.00	NA	NA	0.0849	98.12	1.8800	NA	NA	NA	NA
KUL	DARNet	0.1	Baseline	0.1098	95.45	0.1965	93.10	0.4348	87.28	8.1700	5.8200	-0.3500	-0.1900	0.1100
KUL	DARNet	0.1	PAL	0.1196	95.10	0.1932	92.91	0.4023	87.39	7.7100	5.5200	NA	NA	NA
KUL	DARNet	1	Baseline	0.0281	98.99	0.1092	96.79	0.3457	91.91	7.0800	4.8800	-0.0100	0.1000	0.3100
KUL	DARNet	1	PAL	0.0282	98.98	0.1040	96.89	0.3103	92.22	6.7600	4.6700	NA	NA	NA
KUL	DARNet	2	Baseline	0.0176	99.45	0.1020	96.99	0.3547	92.15	7.3000	4.8400	-0.5200	-0.1400	0.5000
KUL	DARNet	2	PAL	0.0337	98.93	0.1003	96.85	0.2871	92.65	6.2800	4.2000	NA	NA	NA
DTU	SSF-CNN	0.1	Baseline	0.6278	64.65	0.6369	64.07	0.6426	63.92	0.7300	0.1500	5.5800	4.8400	5.3300
DTU	SSF-CNN	0.1	PAL	0.5582	70.23	0.5794	68.91	0.5818	69.25	0.9800	0.3400	NA	NA	NA
DTU	SSF-CNN	1	Baseline	0.5807	68.20	0.6217	64.97	0.6413	65.84	2.3600	0.8700	6.5800	5.5600	5.1100
DTU	SSF-CNN	1	PAL	0.4826	74.78	0.5811	70.53	0.6059	70.95	3.8300	0.4200	NA	NA	NA
DTU	SSF-CNN	2	Baseline	0.5619	69.40	0.6316	64.93	0.6354	67.94	1.4600	3.0100	5.9400	3.7900	4.1500
DTU	SSF-CNN	2	PAL	0.4736	75.34	0.5936	68.72	0.5799	72.09	3.2500	3.3700	NA	NA	NA
DTU	STAnet	0.1	Baseline	0.6401	63.14	NA	NA	0.6689	59.09	4.0500	NA	35.4000	NA	37.9600
DTU	STAnet	0.1	PAL	0.0702	98.54	NA	NA	0.1341	97.05	1.4900	NA	NA	NA	NA
DTU	STAnet	1	Baseline	0.4601	76.52	NA	NA	0.7636	64.12	12.4000	NA	22.9300	NA	32.5900
DTU	STAnet	1	PAL	0.0353	99.45	NA	NA	0.1758	96.71	2.7400	NA	NA	NA	NA
DTU	STAnet	2	Baseline	0.4590	76.00	NA	NA	0.8731	61.59	14.4100	NA	22.2300	NA	32.4400
DTU	STAnet	2	PAL	0.0641	98.23	NA	NA	0.2674	94.03	4.2000	NA	NA	NA	NA
DTU	DBPNet	0.1	Baseline	0.1257	94.94	NA	NA	1.8504	70.72	24.2200	NA	0.9500	NA	7.2500
DTU	DBPNet	0.1	PAL	0.1043	95.89	NA	NA	1.2684	77.97	17.9200	NA	NA	NA	NA
DTU	DBPNet	1	Baseline	0.0371	98.84	NA	NA	1.7956	77.55	21.2900	NA	-0.1300	NA	8.6300
DTU	DBPNet	1	PAL	0.0443	98.71	NA	NA	0.5409	86.18	12.5300	NA	NA	NA	NA
DTU	DBPNet	2	Baseline	0.0274	99.37	NA	NA	0.8583	80.88	18.4900	NA	-0.4300	NA	5.9300
DTU	DBPNet	2	PAL	0.0408	98.94	NA	NA	0.4288	86.81	12.1300	NA	NA	NA	NA
DTU	DARNet	0.1	Baseline	0.1083	95.40	0.1388	94.85	1.5024	73.94	21.4600	20.9100	-0.8100	-0.1900	0.3700
DTU	DARNet	0.1	PAL	0.1281	94.59	0.1432	94.66	1.3180	74.31	20.2800	20.3500	NA	NA	NA
DTU	DARNet	1	Baseline	0.0745	97.07	0.2073	93.21	0.9929	78.90	18.1700	14.3100	1.2700	1.4400	1.4800
DTU	DARNet	1	PAL	0.0446	98.34	0.1744	94.65	1.0007	80.38	17.9600	14.2700	NA	NA	NA
DTU	DARNet	2	Baseline	0.0678	97.51	0.2199	92.97	0.8225	80.59	16.9200	12.3800	0.1900	0.8400	0.6500
DTU	DARNet	2	PAL	0.0636	97.70	0.1931	93.81	0.7849	81.24	16.4600	12.5700	NA	NA	NA

Across these 12 settings, mean training accuracy increased from 81.50% to 89.54%, mean validation accuracy from 78.22% to 86.87%, and mean test accuracy from 73.67% to 81.58%. Table 9 shows that the mean train–test gap changed only from 7.83 to 7.96 percentage points, while the mean absolute validation–test gap decreased from 6.00 to 5.75 points.

For DBPNet, train and test records were available for six dataset–window settings. Mean training accuracy changed from 98.56% to 98.87% (+0.31 points), while mean test accuracy increased from 83.65% to 90.88% (+7.22 points). The mean train–test gap consequently decreased from 14.91 to 8.00 points, and all six test comparisons improved by 5.00–11.11 points. For STAnet, mean training/test accuracy changed from 81.65%/75.34% to 98.02%/94.59%; the respective gains were 16.37 and 19.25 points, and the train–test gap decreased from 6.31 to 3.43 points. The native STAnet and DBPNet procedures do not provide independent validation partitions, so validation results are reported as N/A rather than estimated.

Across the 24 matched native-protocol comparisons, test accuracy increased in every configuration. The partition patterns differ by backbone: DARNet changes little in training and modestly in validation/test; SSF-CNN improves across all three partitions; and DBPNet and STAnet improve more on the

Table 8: Partition-specific PAL gains for DARNet and SSF-CNN. Values are PAL minus baseline accuracy in percentage points; AVR is the mean over the three decision windows.

Dataset	Backbone	Window	Train gain	Valid gain	Test gain
DTU	DARNet	0.1 s	−0.81	−0.19	+0.37
DTU	DARNet	1 s	+1.27	+1.44	+1.48
DTU	DARNet	2 s	+0.19	+0.84	+0.65
DTU	DARNet	AVR	+0.22	+0.70	+0.83
DTU	SSF-CNN	0.1 s	+5.58	+4.84	+5.33
DTU	SSF-CNN	1 s	+6.58	+5.56	+5.11
DTU	SSF-CNN	2 s	+5.94	+3.79	+4.15
DTU	SSF-CNN	AVR	+6.03	+4.73	+4.86
KUL	DARNet	0.1 s	−0.35	−0.19	+0.11
KUL	DARNet	1 s	−0.01	+0.10	+0.31
KUL	DARNet	2 s	−0.52	−0.14	+0.50
KUL	DARNet	AVR	−0.29	−0.08	+0.31
KUL	SSF-CNN	0.1 s	+19.93	+20.35	+19.66
KUL	SSF-CNN	1 s	+27.47	+27.31	+27.47
KUL	SSF-CNN	2 s	+31.19	+30.73	+29.73
KUL	SSF-CNN	AVR	+26.20	+26.13	+25.62

Table 9: Aggregated partition performance over the 12 DARNet/SSF-CNN dataset–window settings.

Condition	Train acc. (%)	Valid acc. (%)	Test acc. (%)	Train–test gap	|Valid–test|
Baseline	81.5000	78.2200	73.6717	7.8283	6.0000
PAL	89.5383	86.8700	81.5775	7.9608	5.7508
PAL–baseline	+8.0383	+7.8700	+7.9058	+0.1325	−0.2492

test set than on the training set. The observed gains therefore cannot be explained solely by an increase in training accuracy.

8 Subject-level performance variability across backbones

We examined the low accuracies of S4, S11, and S12 across the evaluated backbones, and of S9 specifically for SSF-CNN, using the original KUL data, preprocessing outputs, training records, and subject-level results. The checks found no missing data, class imbalance, or numerical abnormality. The remaining evidence points to different representation-specific patterns: S4 retains clearly separable information but is poorly matched to some original channel-space representations; S11 and S12 have small attention-related covariance differences relative to within-class variation; and S9 has especially weak separation in the alpha-power topographic representation used by SSF-CNN.

Table 10 summarizes the descriptive mean accuracy of each KUL subject across the four backbones and three decision windows. These cross-backbone averages are used only to identify broad subject-level patterns, because the native model pipelines differ in preprocessing, feature construction, partitioning, and training control. S11, S12, and S4 have the three lowest baseline averages (71.23%, 75.29%, and 79.78%, respectively). Their responses to PAL are nevertheless different: S4 increases to 93.35% (+13.57 points), S12 to 90.85% (+15.57 points), and S11 to 83.07% (+11.85 points). Thus, S11 remains the most difficult subject after PAL, whereas S4 and S12 move out of the lowest-performing group.

Table 10: Subject-wise mean baseline and PAL accuracy on KUL, descriptively averaged across the four backbones and three decision windows. Gain is PAL minus baseline in percentage points. These averages summarize subject-level patterns and are not intended as a strict cross-backbone comparison.

Subject	Baseline (%)	PAL (%)	Gain (points)
S1	82.84	92.92	+10.08
S2	87.84	91.62	+3.78
S3	88.88	96.35	+7.47
S4	**79.78**	**93.35**	**+13.57**
S5	87.01	94.03	+7.02
S6	83.52	93.68	+10.16
S7	81.19	94.72	+13.53
S8	82.55	96.79	+14.24
S9	81.04	83.69	+2.64
S10	80.43	91.85	+11.42
S11	**71.23**	**83.07**	**+11.85**
S12	**75.29**	**90.85**	**+15.57**
S13	84.41	86.94	+2.53
S14	85.40	94.26	+8.87
S15	84.45	94.18	+9.73
S16	88.06	94.69	+6.63
Overall mean	82.74	92.06	+9.32

S4 illustrates a relatively stable and learnable channel-space mismatch. Its covariance ratio is 1.044 (rank 10 of 16), and its standardized left–right alpha-power topographic separation is 0.125 (rank 15 when rank 1 denotes the lowest separation). Therefore, S4 does not show the unusually weak global covariance separation observed for S11 and S12 or the weak alpha-topographic separation observed for S9. Its large gains, particularly with SSF-CNN and DBPNet, instead indicate that discriminative information was present but was not optimally expressed in the original channel organization. PAL appears to make this information more accessible by applying subject-specific channel remixing before backbone-specific feature extraction.

Table 11: Subject-wise between-/within-class covariance diagnostics across all 16 KUL subjects. The between/within ratio is the between-class covariance distance divided by the within-class covariance variability; rank 1 denotes the lowest ratio.

Subject	Between-class covariance distance	Within-class covariance variability	Between/within ratio	Ratio rank (1=lowest)
S1	6.079	4.985	1.219	14
S2	2.760	2.138	1.291	15
S3	2.685	2.266	1.184	13
S4	5.041	4.828	1.044	10
S5	4.738	4.240	1.118	12
S6	2.873	2.980	0.964	8
S7	5.079	5.944	0.854	5
S8	4.814	6.189	0.778	4
S9	1.652	1.847	0.894	7
S10	3.889	5.762	0.675	1
S11	**4.008**	**5.465**	**0.733**	**3**
S12	**5.767**	**8.201**	**0.703**	**2**
S13	2.926	3.321	0.881	6
S14	6.254	4.207	1.487	16
S15	3.094	3.017	1.025	9
S16	4.002	3.825	1.046	11

The covariance ratio compares the between-class covariance distance with within-class covariance variability. S11 and S12 have ratios of 0.733 and 0.703, the third-lowest and second-lowest values among the 16 subjects. For these subjects, attention-related covariance differences are therefore small relative to trial-to-trial variation. For S11 at 0.1 s, the DARNet baseline training, validation, and test accuracies were 88.44%, 80.96%, and 64.27%; for S12 at 1 s, they were 98.18%, 86.40%, and 72.09%. DBPNet showed a similar pattern of high training accuracy and lower test accuracy. These observations are consistent with within-subject distribution shift, but they do not identify its cause. Head conductivity, electrode contact or placement, attention, and residual artifacts are possible contributors.

S11 and S12 nevertheless retain decodable information. Across the three windows, the STAnet gains are 2.58–12.30 points for S11 and 3.52–8.45 points for S12. With DBPNet, S11 changes from 64.75%, 77.29%, and 77.86% to 99.82%, 94.01%, and 91.43%, while S12 changes from 71.57%, 74.47%, and 72.50% to 99.90%, 99.82%, and 100.00%. These improvements are consistent with PAL making retained discriminative information more accessible through subject-specific channel remixing; they do not imply that the underlying source of nonstationarity has been identified.

Table 12: Subject-wise standardized left–right alpha-power topographic separation for the SSF-CNN input across all 16 KUL subjects. All entries use a 2-s window and 1,564 windows; rank 1 denotes the lowest separation.

Subject	Window length (s)	Number of windows	Standardized L/R alpha-power separation	Separation rank (1=lowest)
S1	2	1564	0.046	6
S2	2	1564	0.091	13
S3	2	1564	0.053	7
S4	2	1564	0.125	15
S5	2	1564	0.109	14
S6	2	1564	0.043	5
S7	2	1564	0.059	9
S8	2	1564	0.028	3
S9	**2**	**1564**	**0.018**	**1**
S10	2	1564	0.037	4
S11	2	1564	0.026	2
S12	2	1564	0.068	10
S13	2	1564	0.084	11
S14	2	1564	0.084	12
S15	2	1564	0.135	16
S16	2	1564	0.058	8

S9 exhibits a different limitation. SSF-CNN restricts the signal to 8–13 Hz, aggregates Fourier-domain alpha power, and projects the result onto a fixed scalp topology. S9 has a standardized left–right alpha-power separation of 0.017526 (0.018 after rounding), the lowest among the 16 subjects. Its training, validation, and test accuracies remain near chance at the 1- and 2-s windows, indicating that this representation is already weakly separable during training. S9 is not globally undecodable: PAL-equipped DBPNet reaches 99.74%, 97.89%, and 98.21%, and PAL-equipped DARNet reaches 90.88%, 95.56%, and 95.69%. PAL also raises the 0.1-s SSF-CNN accuracy from 51.25% to 71.06%, but it cannot restore broadband temporal, phase, or covariance information removed by alpha-only processing at longer windows.

The architecture-specific results follow the information retained by each backbone. STAnet models spatial and temporal attention from relatively broadband EEG; SSF-CNN emphasizes alpha-power topology; DBPNet combines CSP-based temporal and multiband spatial–frequency branches; and DARNet applies fixed CSP before embedding and dual attention. Because PAL precedes these feature extractors, its channel remixing reaches two complementary branches in DBPNet. In DARNet, the fixed CSP projection can attenuate adapted directions, and the later embedding and attention stages may overlap with PAL's function. This interpretation is consistent with the S11 DARNet decrease from 64.27%, 72.24%, and 77.15% to 63.83%, 68.90%, and 73.92%, alongside the S12 increase from 71.97%, 72.09%, and 70.83% to 75.55%, 76.02%, and 77.29%.

Taken together, the results suggest that PAL is most useful when discriminative information remains in the signal but its channel organization is poorly matched to the backbone. Its benefit is smaller when the backbone has discarded relevant information or when trial-to-trial variation dominates. Because the models retain different native preprocessing, feature construction, partitions, and training protocols, absolute cross-backbone accuracies are not a strict model ranking. The primary comparison remains the paired baseline–PAL change within each backbone. In this sense, PAL is a common implementation interface, not a guarantee of equal gains for every subject or architecture.

9 Hyperparameter sensitivity

Hyperparameter sensitivity was evaluated through single-factor controlled experiments on KUL (16 subjects; 0.1, 1, and 2 s windows). Rank r, the PAL regularization coefficient λ_2, learning rate, and weight decay were varied one at a time while the data split, random seed, remaining training settings, and early-stopping rule were held fixed. The tables report subject-level mean test accuracy $\pm$ standard deviation for transparency; the test set was not used to select hyperparameters. Selection used validation performance, except that the SSF-CNN learning rate follows the value specified in its original paper.

9.1 Baseline sensitivity

In the baseline models, PAL is disabled; therefore, r and λ_2 do not enter the forward pass or loss. Differences among those rows represent stochastic training variation rather than sensitivity to active baseline parameters. Learning rate and, where evaluated, backbone weight decay are the relevant baseline diagnostics. Table 13 lists every tested value and the corresponding subject-level mean and standard deviation for each decision window.

Table 13: Complete baseline sensitivity on KUL. Each row reports subject-level mean test accuracy ± standard deviation (%) for the indicated parameter value and decision window.

Backbone	Parameter	Value	0.1 s	1 s	2 s
DARNet	rank	2	87.40 ± 10.14	92.13 ± 8.52	91.88 ± 8.60
	rank	4	87.63 ± 9.73	92.36 ± 8.24	92.19 ± 8.81
	rank	8	87.42 ± 9.91	92.02 ± 8.13	92.45 ± 8.35
	rank	16	87.43 ± 9.90	92.19 ± 8.24	92.49 ± 8.47
	rank	32	87.45 ± 9.79	92.34 ± 8.35	92.30 ± 8.51
	rank	64	87.27 ± 10.03	92.14 ± 8.34	91.80 ± 8.35
	λ_2	0	87.42 ± 9.91	92.30 ± 8.02	92.17 ± 8.38
	λ_2	0.0001	87.41 ± 9.80	92.09 ± 8.32	92.47 ± 8.06
	λ_2	0.001	87.45 ± 9.79	92.02 ± 8.13	92.19 ± 8.81
	λ_2	0.01	87.55 ± 9.60	92.00 ± 8.12	93.43 ± 8.98
	learning rate	0.0001	86.01 ± 10.10	90.81 ± 8.46	90.84 ± 9.71
	learning rate	0.00025	86.88 ± 9.85	92.04 ± 7.47	91.38 ± 9.24
	learning rate	0.0005	87.45 ± 9.79	92.02 ± 8.13	92.19 ± 8.81
	learning rate	0.001	87.96 ± 8.91	92.06 ± 7.49	91.80 ± 9.02
	weight decay	0	87.47 ± 9.28	92.35 ± 6.81	92.25 ± 9.72
	weight decay	0.0001	87.33 ± 9.30	92.70 ± 6.82	91.97 ± 8.55
	weight decay	0.0003	87.45 ± 9.79	92.02 ± 8.13	92.19 ± 8.81
	weight decay	0.001	87.67 ± 8.96	92.87 ± 6.81	91.38 ± 10.10
DBPNet	rank	2	86.33 ± 9.03	92.09 ± 7.64	93.17 ± 8.86
	rank	4	86.21 ± 9.29	93.28 ± 7.39	93.44 ± 6.86
	rank	8	86.45 ± 8.98	92.86 ± 7.03	93.17 ± 7.59
	rank	16	86.67 ± 8.89	92.69 ± 7.85	93.33 ± 7.59
	rank	32	86.64 ± 8.96	93.05 ± 7.61	94.00 ± 6.29
	rank	64	86.50 ± 8.96	93.16 ± 7.24	93.24 ± 7.52
	λ_2	0	86.27 ± 9.22	92.94 ± 7.64	93.55 ± 7.09
	λ_2	0.0001	86.30 ± 9.16	92.87 ± 7.36	93.66 ± 6.84
	λ_2	0.001	86.64 ± 8.96	92.86 ± 7.03	93.44 ± 6.86
	λ_2	0.01	86.57 ± 9.04	92.86 ± 7.86	93.77 ± 6.73
	learning rate	0.001	85.54 ± 10.25	93.03 ± 6.83	93.21 ± 5.66
	learning rate	0.003	86.64 ± 8.96	92.86 ± 7.03	93.44 ± 6.86
	learning rate	0.01	85.04 ± 8.73	92.87 ± 7.04	93.66 ± 7.44
	weight decay	0	86.39 ± 9.35	92.53 ± 7.92	92.90 ± 8.25
	weight decay	0.00001	86.45 ± 8.73	92.72 ± 7.66	93.62 ± 7.14
	weight decay	0.00003	86.64 ± 8.96	92.86 ± 7.03	93.44 ± 6.86
	weight decay	0.0001	86.42 ± 9.06	92.65 ± 7.39	93.64 ± 6.92
STAnet	rank	2	83.18 ± 5.96	92.47 ± 5.29	91.57 ± 5.73
	rank	4	83.18 ± 5.96	92.38 ± 4.87	91.01 ± 6.37
	rank	8	83.18 ± 5.96	92.58 ± 5.50	91.57 ± 5.73
	rank	16	83.18 ± 5.96	92.47 ± 5.29	91.57 ± 5.73
	rank	32	83.18 ± 5.96	92.58 ± 5.50	91.01 ± 6.37

	rank	64	83.18 ± 5.96	92.58 ± 5.50	91.01 ± 6.37
	λ_2	0	83.18 ± 5.96	92.58 ± 5.50	91.01 ± 6.37
	λ_2	0.0001	83.18 ± 5.96	92.58 ± 5.50	91.01 ± 6.37
	λ_2	0.001	83.18 ± 5.96	92.58 ± 5.50	91.01 ± 6.37
	λ_2	0.01	83.18 ± 5.96	92.58 ± 5.50	91.01 ± 6.37
	learning rate	0.0003	82.74 ± 5.75	91.02 ± 5.65	89.92 ± 7.35
	learning rate	0.001	83.18 ± 5.96	92.58 ± 5.50	91.01 ± 6.37
	learning rate	0.003	82.89 ± 6.16	92.91 ± 4.89	92.47 ± 5.12
SSF-CNN	rank	2	56.15 ± 8.37	61.24 ± 8.16	60.59 ± 9.40
	rank	4	56.35 ± 7.90	61.05 ± 7.70	61.70 ± 10.44
	rank	8	56.37 ± 7.80	61.37 ± 8.17	60.96 ± 10.02
	rank	16	55.92 ± 8.13	60.75 ± 8.66	61.35 ± 10.60
	rank	32	55.97 ± 7.86	59.92 ± 8.66	61.28 ± 10.13
	λ_2	0	56.12 ± 8.09	60.84 ± 8.21	61.48 ± 10.07
	λ_2	0.0001	55.43 ± 8.44	61.13 ± 8.12	60.50 ± 9.50
	λ_2	0.001	55.97 ± 7.86	61.37 ± 8.17	61.70 ± 10.44
	λ_2	0.01	56.87 ± 7.29	61.08 ± 8.02	60.48 ± 9.60
	learning rate	0.0003	55.10 ± 9.19	60.33 ± 7.48	61.09 ± 8.74
	learning rate	0.001	55.97 ± 7.86	61.37 ± 8.17	61.70 ± 10.44
	learning rate	0.003	53.27 ± 6.33	55.34 ± 8.98	54.90 ± 7.97

The baseline results distinguish genuine optimization sensitivity from variation in inactive PAL settings. Learning-rate changes, particularly for SSF-CNN, produce larger differences than the inactive rank and λ_2 rows. Reporting every tested value also makes the scale of stochastic variation and the backbone-specific response explicit.

9.2 PAL sensitivity

For PAL models, REST first supplies the standard-head-model reference space and the personalized matrix then performs task-driven channel modulation; r and λ_2 are therefore active. Table 14 reports the complete sweep. DARNet and DBPNet vary little across the tested ranges, and STAnet reaches a broad plateau as rank increases. SSF-CNN improves as rank increases from 2 to 32 but is more sensitive to the learning rate: changing it from 0.001 to 0.003 reduces accuracy substantially. We therefore retain 0.001, the RMSprop setting reported for the original SSF-CNN, rather than selecting a value from test performance.

Table 14: Complete PAL sensitivity on KUL. Each row reports subject-level mean test accuracy ± standard deviation (%) for the indicated parameter value and decision window.

Backbone	Parameter	Value	0.1 s	1 s	2 s
DARNet	rank	2	86.91 ± 9.84	91.91 ± 9.11	91.86 ± 7.82
	rank	4	86.74 ± 9.82	92.39 ± 7.74	92.27 ± 7.72
	rank	8	86.15 ± 9.55	92.00 ± 9.07	92.69 ± 6.85
	rank	16	87.19 ± 9.37	92.32 ± 8.08	93.01 ± 6.76
	rank	32	87.28 ± 9.46	92.15 ± 8.42	92.53 ± 8.11
	rank	64	87.58 ± 9.07	92.09 ± 8.14	93.58 ± 7.06
	λ_2	0	87.29 ± 9.53	91.63 ± 9.09	92.66 ± 7.85
	λ_2	0.0001	87.36 ± 9.57	92.34 ± 8.63	92.23 ± 7.76
	λ_2	0.001	87.28 ± 9.46	92.00 ± 9.07	92.27 ± 7.72
	λ_2	0.01	86.92 ± 9.47	92.42 ± 7.30	92.25 ± 8.08
	learning rate	0.0001	85.41 ± 10.09	90.82 ± 8.35	90.62 ± 8.92
	learning rate	0.00025	86.44 ± 9.52	90.98 ± 8.69	91.25 ± 8.30
	learning rate	0.0005	87.28 ± 9.46	92.00 ± 9.07	92.27 ± 7.72
	learning rate	0.001	87.94 ± 8.94	93.35 ± 7.67	92.32 ± 7.76
	weight decay	0	87.23 ± 9.27	92.68 ± 7.56	92.43 ± 7.41
	weight decay	0.0001	87.21 ± 9.41	92.82 ± 7.15	93.12 ± 7.18
	weight decay	0.0003	87.28 ± 9.46	92.00 ± 9.07	92.27 ± 7.72
	weight decay	0.001	87.03 ± 9.39	92.72 ± 7.27	92.38 ± 8.43
DBPNet	rank	2	96.32 ± 2.91	97.93 ± 2.87	98.26 ± 2.36
	rank	4	97.05 ± 2.83	98.18 ± 2.41	98.04 ± 2.95
	rank	8	97.37 ± 2.85	98.18 ± 2.37	98.64 ± 2.20
	rank	16	97.66 ± 2.72	98.61 ± 1.87	98.42 ± 2.46
	rank	32	97.89 ± 2.63	98.48 ± 2.28	98.59 ± 2.20
	rank	64	97.89 ± 2.70	98.48 ± 2.35	98.79 ± 1.77
	λ_2	0	97.85 ± 2.72	98.14 ± 2.57	98.12 ± 2.57
	λ_2	0.0001	97.81 ± 2.82	98.57 ± 2.09	98.21 ± 2.46
	λ_2	0.001	97.89 ± 2.63	98.18 ± 2.37	98.04 ± 2.95
	λ_2	0.01	97.70 ± 2.69	98.50 ± 2.13	98.17 ± 2.49
	learning rate	0.001	97.36 ± 2.87	98.22 ± 2.41	98.15 ± 2.65
	learning rate	0.003	97.89 ± 2.63	98.18 ± 2.37	98.04 ± 2.95
	learning rate	0.01	97.80 ± 2.92	98.28 ± 2.38	98.39 ± 2.40
	weight decay	0	97.72 ± 2.73	98.31 ± 2.26	98.08 ± 2.49
	weight decay	0.00001	97.85 ± 2.67	98.47 ± 2.07	98.35 ± 2.33
	weight decay	0.00003	97.89 ± 2.63	98.18 ± 2.37	98.04 ± 2.95
	weight decay	0.0001	97.66 ± 2.83	98.26 ± 2.32	97.83 ± 2.82
STAnet	rank	2	89.12 ± 5.15	94.57 ± 3.64	93.36 ± 4.21
	rank	4	90.65 ± 4.72	94.94 ± 3.50	93.31 ± 4.30
	rank	8	91.60 ± 4.29	94.82 ± 3.40	93.40 ± 4.20
	rank	16	91.90 ± 4.22	95.08 ± 2.82	93.97 ± 3.78
	rank	32	92.11 ± 4.15	94.83 ± 3.33	93.95 ± 3.77
	rank	64	91.93 ± 4.39	94.96 ± 3.52	94.19 ± 3.55

	λ_2	0	92.56 ± 4.31	95.23 ± 3.02	93.14 ± 3.70
	λ_2	0.0001	92.21 ± 4.25	95.15 ± 2.98	92.88 ± 4.11
	λ_2	0.001	92.11 ± 4.15	94.82 ± 3.40	93.31 ± 4.30
	λ_2	0.01	91.50 ± 4.51	94.56 ± 3.39	92.66 ± 4.17
	learning rate	0.0003	91.33 ± 3.96	93.48 ± 3.39	90.85 ± 4.66
	learning rate	0.001	92.11 ± 4.15	94.82 ± 3.40	93.31 ± 4.30
	learning rate	0.003	92.46 ± 4.59	96.39 ± 2.85	95.49 ± 3.89
SSF-CNN	rank	2	72.95 ± 13.04	80.19 ± 15.66	83.77 ± 13.06
	rank	4	76.25 ± 13.30	81.89 ± 15.43	86.33 ± 14.07
	rank	8	77.39 ± 13.00	82.89 ± 15.15	89.32 ± 8.94
	rank	16	76.86 ± 14.57	86.65 ± 11.13	91.62 ± 7.59
	rank	32	78.99 ± 12.46	88.29 ± 10.10	92.86 ± 6.49
	λ_2	0	77.05 ± 14.38	85.68 ± 12.87	84.79 ± 13.55
	λ_2	0.0001	78.73 ± 12.88	84.86 ± 12.63	84.79 ± 14.12
	λ_2	0.001	78.99 ± 12.46	82.89 ± 15.15	86.33 ± 14.07
	λ_2	0.01	78.26 ± 12.83	81.94 ± 15.23	84.83 ± 12.41
	learning rate	0.0003	76.88 ± 13.43	80.81 ± 14.65	77.71 ± 14.19
	learning rate	0.001	78.99 ± 12.46	82.89 ± 15.15	86.33 ± 14.07
	learning rate	0.003	68.56 ± 19.90	71.42 ± 19.69	76.30 ± 20.73

All PAL parameters are optimized jointly with the corresponding backbone under the stated optimizer and schedule; no separate optimization stage or auxiliary feature-extraction network is used. Most sweeps show broad accuracy plateaus, so the reported gains are not tied to a single isolated setting. The stronger SSF-CNN sensitivity also shows why the same rank or optimization setting should not be assumed optimal for every backbone.

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